Lending Operations Data Pipeline

Dataset documentation — sanitized sample, portfolio demonstration

Context

This document accompanies a sanitized data artifact published alongside the “Conversational AI for Lending Operations” case study. The artifact represents the data layer built for an internal AI-driven operations system for a Kenyan lending / microfinance business: the same pipeline that feeds the system's natural-language agent, dashboards, and automated reminder/escalation engine.

Pipeline

At a high level, data moves through four stages:

- **Source system** — Microsoft Dynamics 365 Business Central, the business's ERP, holding loan, client, staff, and branch records.
- **Extraction & synchronization** — scheduled sync jobs pull records from the ERP and upsert them into an application-owned database, replacing what was previously a manual, hand-exported CSV workflow.
- **Normalized PostgreSQL layer** — records land in related tables (clients, loans, staff, branches, repayment schedules) rather than flat exports, plus derived views computed on top of them (for example, which loans are overdue and by how much).
- **Downstream access** — the same normalized data is queried by Power BI dashboards, by a natural-language-to-SQL AI agent, and by an automated engine that sends reminders and escalates unresolved issues.

Dataset

The published sample models the **loan portfolio layer** of the system — the shape of a loan record after it has been synced and enriched with a derived operational field, matching the structure of the production *loans* table and its *overdue* view.

60 rows, 15 columns. This is a small representative sample built for demonstration, not a production export or a claim about actual portfolio size.

Field	Description
loan_ref	Synthetic loan identifier (sample-only, not a real loan ID format)
branch_ref	Synthetic branch code
loan_status	One of: pending, active, overdue, closed, cancelled, defaulted, written_off — the real status values used in the production schema
repayment_cycle	One of: 4 weeks, 6 weeks, 8 weeks, 12 weeks — the real repayment cycle options used in the production schema
installments	Number of installments for the loan's repayment cycle

Field	Description
principal_amount / interest_amount / total_amount	Loan principal, interest, and total amount (synthetic figures)
total_paid / outstanding_balance / arrears_amount	Running payment state for the loan (synthetic figures)
issued_date / expected_completion_date / actual_completion_date	Loan lifecycle dates (synthetic)
overdue_days	Derived field — mirrors how the production overdue view computes days overdue from the earliest unpaid installment

The production *clients* table also includes identity fields — full name, national ID, phone, email. Those fields are described in the Privacy Note below and are not present anywhere in this sample.

Example Data Model

Confirmed relationships in the production schema, shown at a high level:

```
Clients → Loans → Repayment Schedule / Repayments → derived operational views (overdue, arrears)
→ reporting, conversational AI access, and automated reminders / escalation
```

Loans also reference the staff member and branch responsible for them, and every reminder, escalation, payment, or follow-up note is recorded against a loan as a typed event, so automated and human actions can both be traced.

Transformations

- **Synchronization** — scheduled jobs pull ERP records and upsert them into the application's own database on a recurring basis.
- **Normalization** — records that were previously exported by hand as flat CSVs are restructured into related tables (clients, loans, staff, branches) with foreign-key relationships.
- **Type conversion & constraints** — monetary fields are stored as constrained numeric types (non-negative checks), and status fields are restricted to a fixed set of enumerated values rather than free text.
- **Derived fields** — fields such as days overdue are computed from underlying repayment-schedule data rather than stored directly.
- **Joining** — operational views join loans with clients, staff, and repayment-schedule records to directly answer questions like “which loans are overdue, and whose are they.”
- **Aggregation** — portfolio-level performance figures (such as portfolio-at-risk and collection rate) are computed across a manager's or branch's loan book on top of this same layer.

What This Demonstrates

- Data ingestion from a real external ERP system into an application-owned database.
- Relational data modeling that reflects real business relationships, not a flat export.
- Derived, computed views that turn raw records into directly answerable operational questions.

- Making operational data accessible beyond a fixed report — through dashboards and a natural-language interface.
- Feeding structured, verified data into downstream automation, rather than acting on unverified guesses.

Privacy Note

This dataset has been sanitized for portfolio demonstration. It does not contain real customer names, national IDs, phone numbers, email addresses, or any other production customer data — those fields exist in the production schema but have been **excluded entirely** from this sample, not merely masked. All values shown (loan amounts, dates, statuses, branch codes) are synthetically generated and do not reflect actual loans, clients, or business performance.
